

Project

Q1: List the names and departments of students who have more than 85% attendance and are skilled in both "MongoDB" and "Python".

Query:

```
// Query 1 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  name: 'Steven Wong',
  department: 'Biotechnology'
}
```

Q2: Show all faculty who are teaching more than 2 courses. Display their names and the total number of courses they teach.

Query:

```
>_MONGOSH

// Query 2 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  _id: 'F029',
  name: 'Charles Newton',
  totalCourses: 3
}
{
  _id: 'F083',
  name: 'Matthew Hanna',
  totalCourses: 3
}
{
  _id: 'F084',
  name: 'Michael Johnson',
  totalCourses: 3
}
```

Q3: Write a query to show each student's name along with the course titles they are enrolled in (use \$lookup between enrollments, students, and courses).

Query:

```
>_MONGOSH
```

```
// Query 3 | Name: Omi Kumar | University Roll No.: 1240258308
```

```
< {  
  student_name: 'Kathryn Ferguson',  
  course_title: 'Monitored bottom-line capability'  
}  
{  
  student_name: 'Elizabeth Reed',  
  course_title: 'Intuitive actuating superstructure'  
}  
{  
  student_name: 'Gabriela Le',  
  course_title: 'Optional neutral workforce'  
}
```

Q4: For each course, display the course title, number of students enrolled, and average marks (use \$group).

Query:

```
>_MONGOSH
```

```
}}
```

```
// Query 4 | Name: Omi Kumar | University Roll No.: 1240258308
```

```
< {
```

```
  studentCount: 2,
```

```
  avgMarks: 67.5,
```

```
  course_title: 'Streamlined zero administration strategy'
```

```
}
```

```
{
```

```
  studentCount: 1,
```

```
  avgMarks: 81,
```

```
  course_title: 'User-centric upward-trending functionalities'
```

```
}
```

```
{
```

```
  studentCount: 2,
```

```
  avgMarks: 75.5,
```

```
  course_title: 'Focused client-server knowledge user'
```

```
}
```

Q5: Find the top 3 students with the highest average marks across all enrolled courses.

Query:

```
// Query 5 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  avgMarks: 100,
  name: 'Diane Phillips'
}
{
  avgMarks: 98,
  name: 'Brandon Rios'
}
{
  avgMarks: 94,
  name: 'Larry Ramsey'
}
```

Q6: Count how many students are in each department. Display the department with the highest number of students.

Query:

```
// Query 6 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  studentCount: 23,
  department: 'Electrical'
}
{
  studentCount: 22,
  department: 'Civil'
}
{
  studentCount: 20,
  department: 'Computer Science'
}
```

Q7: Update attendance to 100% for all students who won any "Hackathon".

Query:

```
// Query 7 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 9,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Q8: Delete all student activity records where the activity year is before 2022.

Query:

```
// Query 8 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  acknowledged: true,
  deletedCount: 0
}
```

Q9 Upsert a course record for "Data Structures" with ID "C150" and credits 4—if it doesn't exist, insert it; otherwise update its title to "Advanced Data Structures".

Query:

```
// Query 9 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  acknowledged: true,
  insertedId: 'C150',
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 1
}
```

Q10: Find all students who have "Python" as a skill but not "C++"

Query:

```
>_MONGOSH
// Query 10 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  name: 'Thomas Jackson',
  skills: [
    'Python',
    'AutoCAD'
  ]
}
{
  name: 'Kyle Hale',
  skills: [
    'Python',
    'Java'
  ]
}
{
  name: 'Derrick Humphrey',
  skills: [
    'Python',
    'Java'
  ]
}
```

Q11: Return names of students who participated in "Seminar" and "Hackathon" both.

Query:

```
// Query 11 | Name: Omi Kumar | University Roll No.: 1240258308
< {
  name: 'Lydia Day'
}
{
  name: 'Carlos Bryant'
}
{
  name: 'Adam Solomon'
}
{
  name: 'Patricia Scott'
}
{
  name: 'Taylor Webb'
}
```

Q12: Find students who scored more than 80 in "Web Development" only if they belong to the "Computer Science" department.

Query:

```
// Query 12 | Name: Omi Kumar | University Roll No.: 1240258308  
<
```

Q13: For each faculty member, list the names of all students enrolled in their courses along with average marks per student per faculty.

Query:

```
>_MONGOSH
```

```
//
```

```
// Query 13 | Name: Omi Kumar | University Roll No.: 1240258308
```

```
< {  
  avgMarks: 51,  
  faculty_name: 'Michael Johnson',  
  student_name: 'Carolyn Chandler'  
}  
{  
  avgMarks: 90,  
  faculty_name: 'William Alexander',  
  student_name: 'Vincent Norris'  
}  
{  
  avgMarks: 51,  
  faculty_name: 'Charles Cunningham',  
  student_name: 'Carolyn Chandler'  
}
```

Q14: Show the most popular activity type (e.g., Hackathon, Seminar, etc.) by number of student participants.

Query:

```
//
```

```
// Query 14 | Name: Omi Kumar | University Roll No.: 1240258308
```

```
< {  
  _id: 'Seminar',  
  participants: 35  
}
```